FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: multisource
Model status: BOUNDED_MULTI_SOURCE_3PH_PHASOR
Captured UTC: 2026-10-26T06:04:33.395Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: nominalLLV | Entered value: 13800
Input path: voltageFactor | Entered value: 1.1
Input path: frequencyHz | Entered value: 60
Input path: faultBus | Entered value: F
Input path: buses[0] | Entered value: F
Input path: buses[1] | Entered value: A
Input path: buses[2] | Entered value: B
Input path: branches[0].id | Entered value: feeder-A
Input path: branches[0].from | Entered value: A
Input path: branches[0].to | Entered value: F
Input path: branches[0].rOhm | Entered value: 0.06
Input path: branches[0].xOhm | Entered value: 0.12
Input path: branches[1].id | Entered value: feeder-B
Input path: branches[1].from | Entered value: B
Input path: branches[1].to | Entered value: F
Input path: branches[1].rOhm | Entered value: 0.04
Input path: branches[1].xOhm | Entered value: 0.13
Input path: sources[0].id | Entered value: utility-1
Input path: sources[0].bus | Entered value: A
Input path: sources[0].rOhm | Entered value: 0.08
Input path: sources[0].xOhm | Entered value: 0.42
Input path: sources[0].angleDeg | Entered value: 0
Input path: sources[0].emScale | Entered value: 1
Input path: sources[1].id | Entered value: generator-2
Input path: sources[1].bus | Entered value: B
Input path: sources[1].rOhm | Entered value: 0.11
Input path: sources[1].xOhm | Entered value: 0.6
Input path: sources[1].angleDeg | Entered value: 0
Input path: sources[1].emScale | Entered value: 1

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: BOUNDED_MULTI_SOURCE_3PH_PHASOR
Calculated output path: method | Value: Balanced three-phase bolted or explicitly specified finite R+jX fault impedance, from fundamental-frequency independent Thevenin voltage sources at one voltage level. Direct nodal current law. Not complete IEC 60909:2026 or breaker duty study.
Calculated output path: nominalLLV | Value: 13800
Calculated output path: voltageFactor | Value: 1.1
Calculated output path: frequencyHz | Value: 60
Calculated output path: faultBus | Value: F
Calculated output path: phaseFaultCurrentA.real | Value: 6309.743758583885
Calculated output path: phaseFaultCurrentA.imag | Value: -26727.116406823567
Calculated output path: phaseFaultCurrentA.magnitude | Value: 27461.82109261667
Calculated output path: phaseFaultCurrentA.angleDegrees | Value: -76.7168095145296
Calculated output path: initialSymmetricalKA | Value: 27.461821092616667

Calculated output path: initialShortCircuitMVA | Value: 656.4007177307465
Calculated output path: sourceContributions[0].id | Value: utility-1
Calculated output path: sourceContributions[0].bus | Value: A
Calculated output path: sourceContributions[0].emfPhaseV.real | Value: 8764.177086298521
Calculated output path: sourceContributions[0].emfPhaseV.imag | Value: 0
Calculated output path: sourceContributions[0].emfPhaseV.magnitude | Value: 8764.177086298521
Calculated output path: sourceContributions[0].emfPhaseV.angleDegrees | Value: 0
Calculated output path: sourceContributions[0].currentA.real | Value: 3942.7531879235003
Calculated output path: sourceContributions[0].currentA.imag | Value: -15207.762296276358
Calculated output path: sourceContributions[0].currentA.magnitude | Value: 15710.54858243103
Calculated output path: sourceContributions[0].currentA.angleDegrees | Value: -75.46554491945989
Calculated output path: sourceContributions[1].id | Value: generator-2
Calculated output path: sourceContributions[1].bus | Value: B
Calculated output path: sourceContributions[1].emfPhaseV.real | Value: 8764.177086298521
Calculated output path: sourceContributions[1].emfPhaseV.imag | Value: 0
Calculated output path: sourceContributions[1].emfPhaseV.magnitude | Value: 8764.177086298521
Calculated output path: sourceContributions[1].emfPhaseV.angleDegrees | Value: 0
Calculated output path: sourceContributions[1].currentA.real | Value: 2366.990570660386
Calculated output path: sourceContributions[1].currentA.imag | Value: -11519.35411054721
Calculated output path: sourceContributions[1].currentA.magnitude | Value: 11760.023957704172
Calculated output path: sourceContributions[1].currentA.angleDegrees | Value: -78.3885135761115
Calculated output path: branchCurrents[0].id | Value: feeder-A
Calculated output path: branchCurrents[0].from | Value: A
Calculated output path: branchCurrents[0].to | Value: F
Calculated output path: branchCurrents[0].currentA.real | Value: 3942.7531879234994
Calculated output path: branchCurrents[0].currentA.imag | Value: -15207.762296276356
Calculated output path: branchCurrents[0].currentA.magnitude | Value: 15710.548582431029
Calculated output path: branchCurrents[0].currentA.angleDegrees | Value: -75.46554491945989
Calculated output path: branchCurrents[1].id | Value: feeder-B
Calculated output path: branchCurrents[1].from | Value: B
Calculated output path: branchCurrents[1].to | Value: F
Calculated output path: branchCurrents[1].currentA.real | Value: 2366.9905706603854
Calculated output path: branchCurrents[1].currentA.imag | Value: -11519.354110547209
Calculated output path: branchCurrents[1].currentA.magnitude | Value: 11760.02395770417
Calculated output path: branchCurrents[1].currentA.angleDegrees | Value: -78.3885135761115
Calculated output path: busVoltages[0].bus | Value: F
Calculated output path: busVoltages[0].phaseVoltageV.real | Value: 0
Calculated output path: busVoltages[0].phaseVoltageV.imag | Value: 0
Calculated output path: busVoltages[0].phaseVoltageV.magnitude | Value: 0
Calculated output path: busVoltages[0].phaseVoltageV.angleDegrees | Value: 0
Calculated output path: busVoltages[1].bus | Value: A
Calculated output path: busVoltages[1].phaseVoltageV.real | Value: 2061.496666828573
Calculated output path: busVoltages[1].phaseVoltageV.imag | Value: -439.3353552257615
Calculated output path: busVoltages[1].phaseVoltageV.magnitude | Value: 2107.7912756477244
Calculated output path: busVoltages[1].phaseVoltageV.angleDegrees | Value: -12.03059609653787
Calculated output path: busVoltages[2].bus | Value: B
Calculated output path: busVoltages[2].phaseVoltageV.real | Value: 1592.1956571975527
Calculated output path: busVoltages[2].phaseVoltageV.imag | Value: -153.06539023603824
Calculated output path: busVoltages[2].phaseVoltageV.magnitude | Value: 1599.5361904273555
Calculated output path: busVoltages[2].phaseVoltageV.angleDegrees | Value: -5.491242545163879
Calculated output path: faultMode | Value: BOLTED_THREE_PHASE
Calculated output path: faultImpedanceOhm.rOhm | Value: 0
Calculated output path: faultImpedanceOhm.xOhm | Value: 0
Calculated output path: faultBusVoltageV.real | Value: 0
Calculated output path: faultBusVoltageV.imag | Value: 0
Calculated output path: faultBusVoltageV.magnitude | Value: 0
Calculated output path: faultBusVoltageV.angleDegrees | Value: 0
Calculated output path: nodalRelativeResidual | Value: 9.146850691977855e-17
Calculated output path: kclRelativeResidual | Value: 3.311851383437398e-17

Calculated output path: limits[0] | Value: One nominal voltage level: transformer turns ratio and vector group NOT modelled
Calculated output path: limits[1] | Value: Sources are independent fixed Thevenin EMFs with entered actual positive-sequence Z
Calculated output path: limits[2] | Value: No generator or inverter decrement, motor fault contributions unless represented by documented source impedances
Calculated output path: limits[3] | Value: No making peak, DC decrement or breaking duty calculated
Calculated output path: limits[4] | Value: Only balanced 3-phase metallic faults with optional linear R+jX fault impedance; no nonlinear arcs or negative-/zero-sequence faults
Calculated output path: limits[5] | Value: No automatic IEC equipment correction factors, utility maximum/minimum source envelope or source impedance qualification

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: multisource
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:33.395Z